Introduction

Team Name: ASHKARA

Name of Team Lead: MHASAWADE PRATIK UTTAM

Name of the Institution: Vidya Pratishthan's Kamalnayan Bajaj Institute of Engineering and Technology, Baramati

Name of the processor: SHAKTI

Proposal Overview I

Title of the proposal:
AI-Powered Drone System for Early Detection of Stampede and Crowd Panic Events

Target Application: Security & Surveillance, Emerging Technologies (Edge AI)

Objectives of the proposal:

1. To design and implement a **SoC on an Artix-7 FPGA** featuring a dual-core SHAKTI processor.
2. To implement an **AMP** (Asymmetric Multiprocessing) system, dedicating one core to flight control and the other to AI inference.
3. To develop an **autonomous drone** capable of real-time video capture and on-board AI processing.
4. To create a low-power, high-performance **indigenous surveillance system** for crowd management

Proposal Overview II

Novelty of the proposal:

- **First indigenous drone** built on **SHAKTI RISC-V SoC + FPGA**.
- **Onboard edge AI** for real-time crowd anomaly detection with **no cloud dependency**.
- **Reprogrammable & modular** FPGA-based flight controller for multiple missions.
- **Dual-core architecture:** Core 0 (flight) + Core 1 (AI) for low-latency response.
- **Open-source & secure** – ideal for Smart City, Homeland Security, and defense.
- **Cost-effective**: Delivers DJI-class capabilities at a fraction of the cost with local manufacturing.

Proposed Approach & Technical Details I

1. Hardware Approach: Custom System-on-Chip (SoC)

- We'll synthesize a custom SoC on an **Artix-7 100T FPGA**.
- The SoC uses the indigenous, open-source **SHAKTI 'Yamuna' Dual-Core RISC-V processor** (RV32IMAC).
- A high-speed **AXI4 Crossbar Interconnect** links processors, memory, and peripherals.

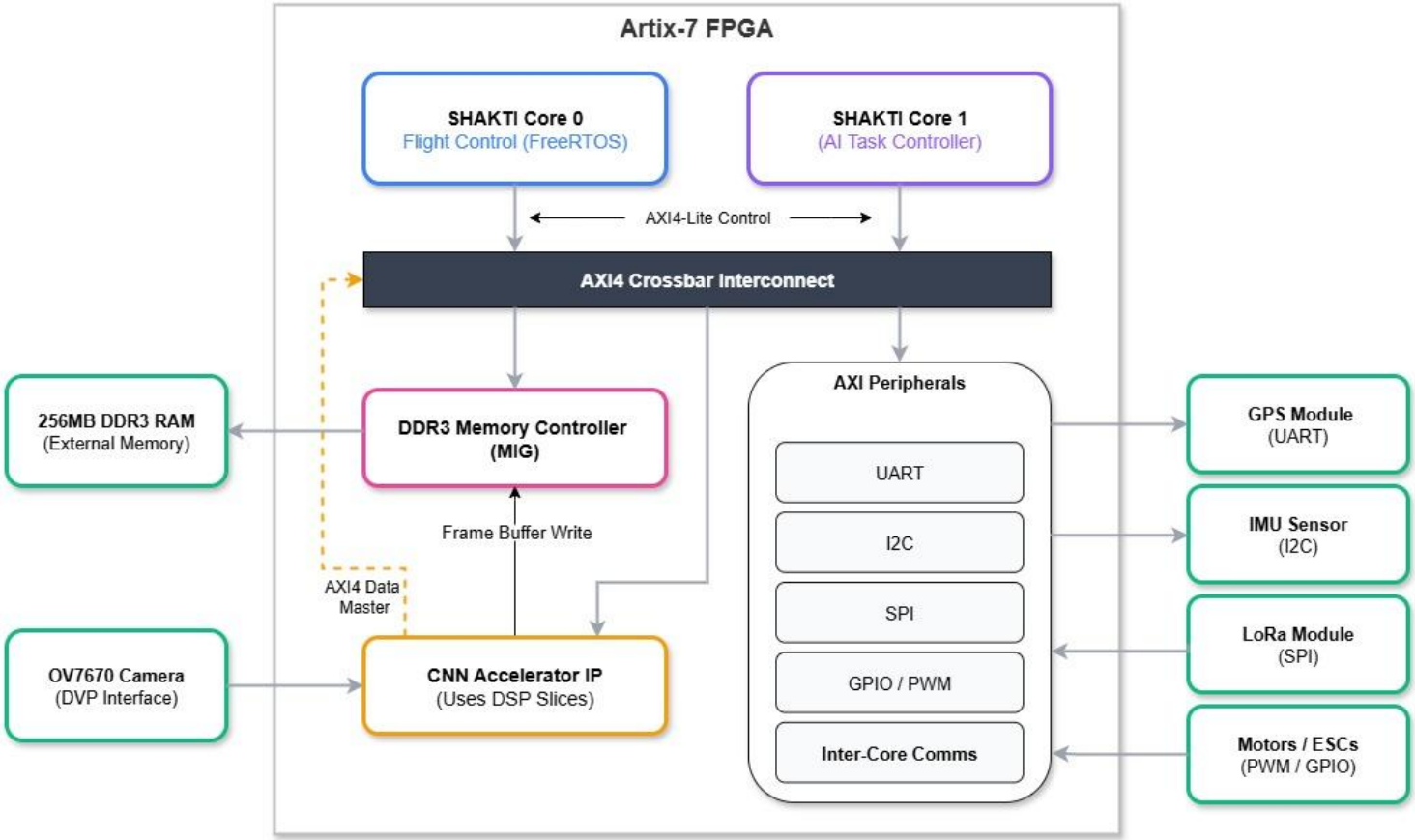
2. Software Architecture: Asymmetric Multiprocessing (AMP) for Safety & Performance

- **Core 0 (Flight Control):** Runs the **FreeRTOS** real-time operating system.
- **Core 1 (AI Controller):** Runs a lightweight, **bare-metal C application**.

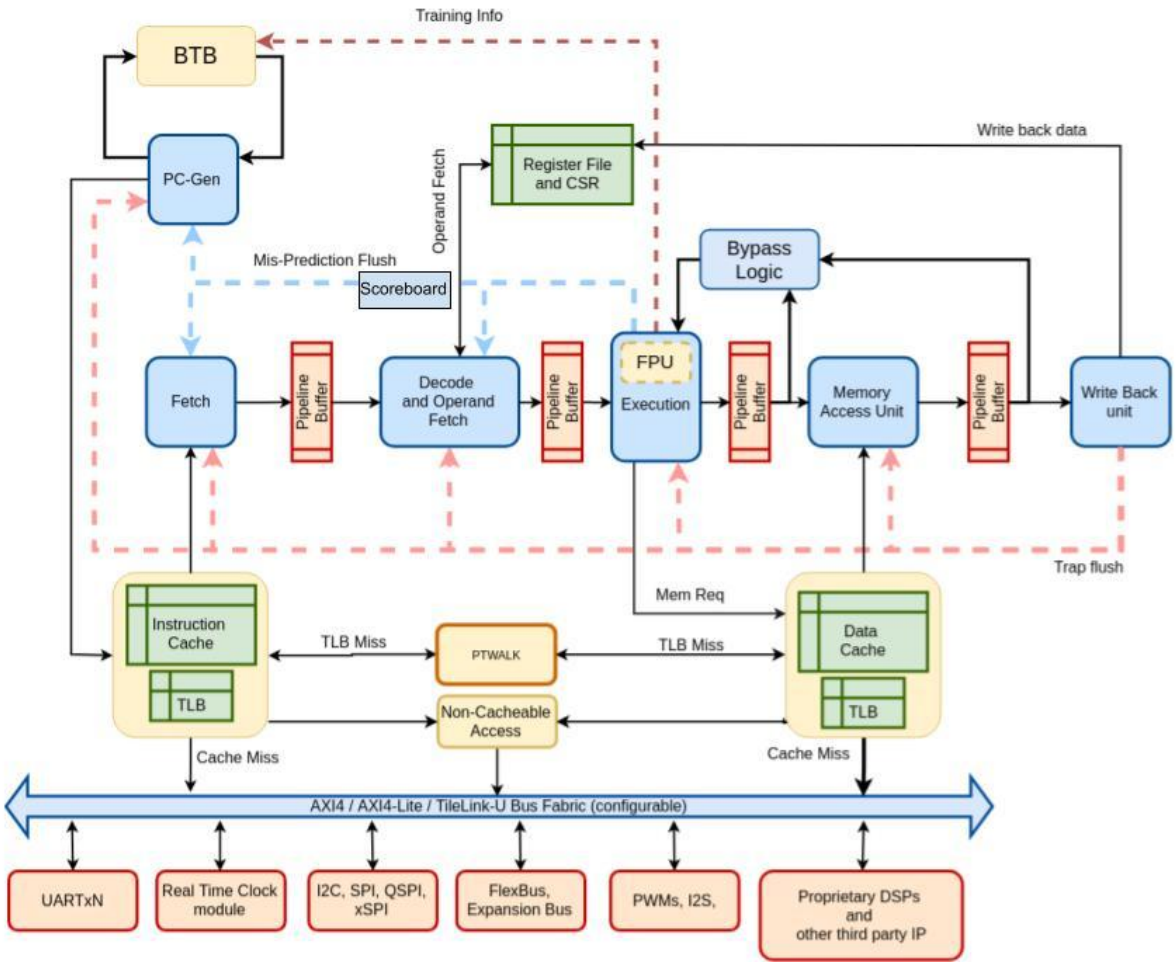
3. AI Acceleration: Hardware/Software Co-Design:

- **AI Model:** A lightweight, 8-bit quantized CNN optimized for 64x64 grayscale video input to minimize memory and computation.
- The **CNN Accelerator** utilizes **DSP slices** for parallel calculations, reducing AI inference latency to 5 ms.

Proposed Approach & Technical Details II (Block Diagram)



Architecture- Shakti Processor



Milestones - Semifinals Stage (MVP & PoC)

Month 1	**Curate & preprocess** crowd datasets **Benchmark lightweight CNNs** (MobileNet Lite, TinyYOLO) **Simulate quantized models** using TensorFlow Lite **Synthesize dual-core Yamuna SoC** and run initial RTL simulations
Month 2	Implement & validate **peripheral IPs** (PWM, UART, I²C, GPIO) in Vivado Establish AXI-based inter-core mailboxes **Port & boot FreeRTOS** on Core 0 Develop **basic testbenches** to ensure timing closure
Month 3	Implement **dummy PID loop** in FreeRTOS(Core 0) Begin **camera interfacing** with OV7670 Test **dummy CNN execution** on Core 1 for baseline inference latency Validate **LoRa alert simulation**

Milestones - Final Stage

Month 4	**Design & 3D-print** the drone frame Select **power-efficient ESCs, motors, and battery** (power budget analysis) Mount **Arty A7-100T board** and integrate **IMU/GPS modules** Validate **sensor fusion** using a Kalman filter
Month 5	Stream **real-time video** to Core 1 using optimized DMA transfers Map **quantized CNN** to a soft-IP accelerator for higher throughput Perform **PID tuning** via hardware-in-the-loop simulation Conduct **manual hover tests** to validate flight stability
Month 6	Integrate **LoRa modem with encryption** Complete **end-to-end stampede detection demo** under simulated scenarios Conduct **autonomous test flights** to validate detection accuracy & latency Finalize **technical documentation & user manual** for submission

Business Potential I

*India's rapidly growing drone market, backed by a thriving **startup ecosystem**, global top-tier ambitions, and a strong push for **self-reliance**, is driving the nation toward becoming a global drone innovation hub*

Target Markets

- **Public safety agencies:** Crowd control and law enforcement; **100+ major city deployments possible.**
- **Industrial site monitoring:** Construction & mining estimated **$5 billion annual global spend.**
- **Critical infrastructure security:** Power plants, pipelines, and borders; India has **4,000+ vulnerable sites.**
- **Large-scale event management:** Over **3,000 high-density gatherings annually** in India alone.

Key Differentiators

- **Low Power:** 40–60% more power-efficient than GPU drones **25–30% longer flight time**.
- **Secure:** 100% auditable **open-source RISC-V platform**, ideal for government use.
- **Cost-Effective: 30–40% lower cost** than ASIC/GPU-based solutions.
- **Adaptable:** Reconfigurable AI & hardware for new missions within **48 hours**.

Implementation Strategy II

Phase	Objective	Facts & Figures
Competition	Build a fully functional prototype and validate via **DIR-V Grand Challenge**	Selected in Top 30; **₹3L milestone funding**
Seed & Partnerships	Raise seed funding and run pilot programs with security agencies	Target **₹50–75L seed funding**, run **3–5 pilot programs**
Productization	Develop ruggedized units, ground control software, and manufacturing pipeline	**50–100 units/year** production; scalable service/support
Market Entry	Launch in niche security & industrial markets	Address **$5B+ market**, focusing on high-value early adopters
Service Expansion	Launch **Drone-as-a-Service (DrAAS)** model with analytics and training support	Offer real-time dashboards, updates, and **RPTO operator training**

Resources at VPKBIET

1. Vivado and Vtis

2. Ongoing Project initiatives with **NIT Goa**

3. Ongoing initiatives to tie up with **K. K. Wagh Nashik** and **MKCL** Maharashtra to provide solutions for Upcoming **Kumbh Mela 2027**

SHAKTI

Home

YAMUNA

A 32-bit C-class SoC on Artix7 100T board

Features:

- C-class 6 stage In-order core
- RV32IMAC supported
- 256 MB of DDR and 8 KB of Boot ROM
- Arduino Compatible pin assignments
- Pin Mux supported
- 32 GPIOs
- 6 PWM controllers
- 4 GPT controllers
- 2 SPI controllers
- 3 UART controllers
- 2 I2C controllers
- One Interrupt controller (PLIC)
- CLINT
- On board XADC
- Onboard FTDI based Debugger
- Ethernet Lite IP from Xilinx

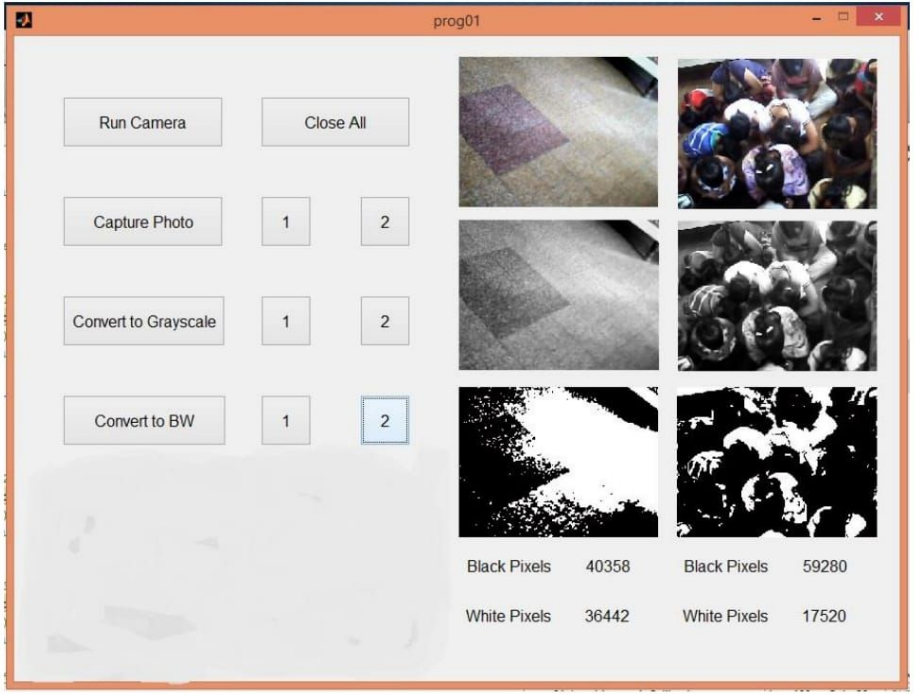
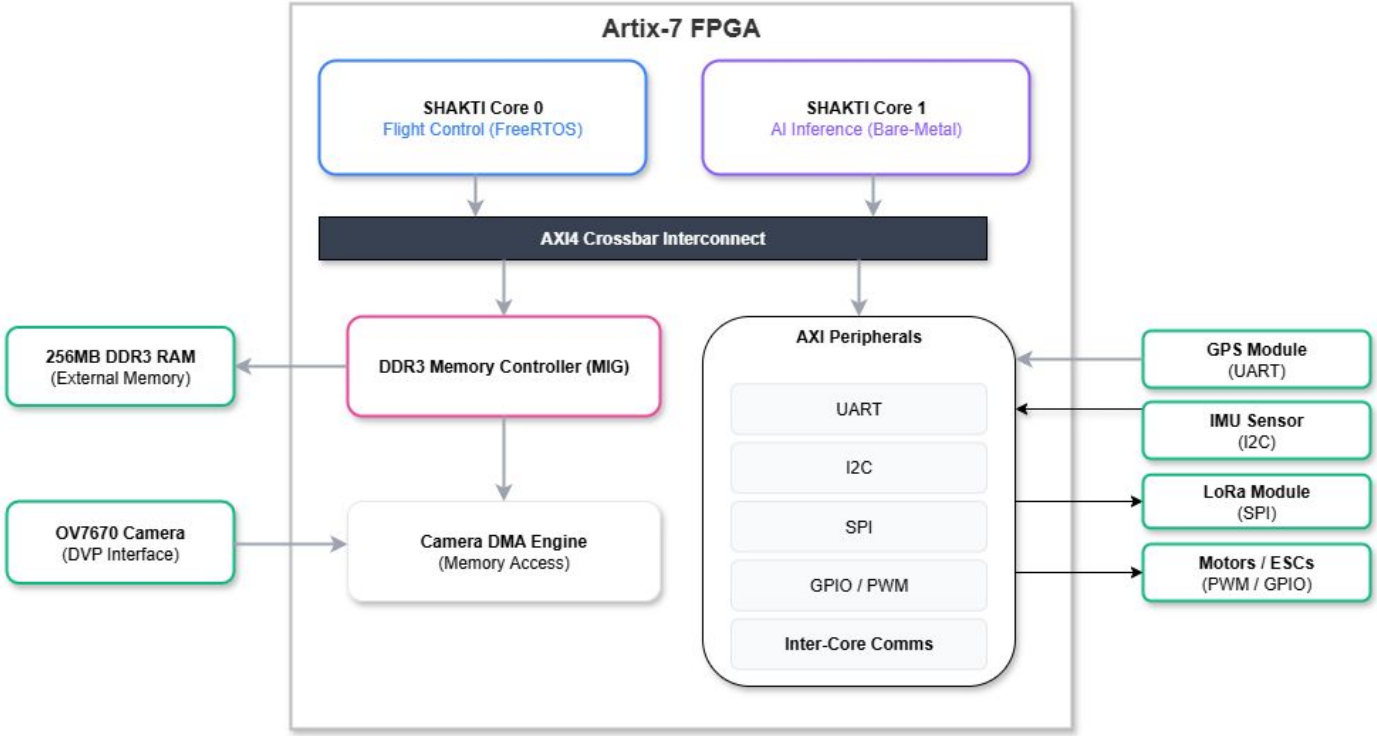


Figure 3. GUI of stampede detection

Proposed Approach & Technical Details II



Block Diagram

Thank You!

Team Ashkara